Spectral-Spatial Deep Support Vector Data Description for Hyperspectral Anomaly Detection

Kun Li, Qiang Ling, Yao Qin, Yingqian Wang, Yaoming Cai, Zaiping Lin, and Wei An

***Abstract*— Hyperspectral anomaly detection (HAD) aims to distinguish anomalies from background-by-background modeling. Deep learning has been applied to HAD and achieves promising detection results. However, there exist several issues that need to be addressed: 1) unrealistic Gaussian assumption on the latent representations may limit its application; 2) deep features are not well-suited to anomaly detection due to the separation between feature learning and anomaly detection; 3) lack of adequate exploitation of spectral-spatial features; 4) negative effect caused by spectral band redundancy. In this article, we propose an end-to-end trainable deep one-class classification network for HAD. Specifically, a minimal enclosing hypersphere is trained to involve the deep features of background samples. These background samples are selected by a density clustering-based method. In this way, feature learning and anomaly detection are incorporated into a unified framework. Meanwhile, there is no explicit Gaussian assumption on the background features. Moreover, due to the complementarity of spectral and spatial features, a novel feature fusion strategy is proposed to fuse spectral and spatial features extracted by a two-stream deep convolutional autoencoder network. Finally, a band attention module is used to automatically learn small weights for redundant bands and thus reduce the negative effect caused by redundant bands. Experimental results on five public datasets demonstrate the superiority of the proposed method compared to several state-of-the-art HAD methods in the detection performance.**

***Index Terms*— Anomaly detection, deep support vector data description (DSVDD), hyperspectral imagery (HSI), one-class classification.**

I. Introduction

HYPERSPECTRAL imagery (HSI) records rich spectral information with hundreds of continuous electromagnetic spectral bands, which can distinguish subtle differences between various materials. Due to this discriminative spectral characteristic, HSI has been widely applied to military and civilian fields [1]–[4]. Among these applications, hyperspectral target detection is a fundamental technique, which aims to separate targets from the background. According to whether the prior spectral information of targets is available, target detection can be categorized into matching target detection and anomaly detection [5]. The former identifies targets based on known spectral signatures of the desired targets. However, the target spectra are difficult to collect in practice, which hinders the applications of matching target detection. In contrast, anomaly detection has drawn much attention since it can identify abnormal targets in an unsupervised manner.

In hyperspectral anomaly detection (HAD), anomalies are identified as the pixels whose spectral signatures are significantly different from their neighboring background clutter pixels [1], [6]. Generally, HAD is a challenging task that lies in twofold: 1) mixed pixels and spectral variability are ubiquitous in HSI data due to complex imaging conditions, which weakens the spectral differences between background and anomalies, and 2) prior knowledge of the target spectra is unavailable. Therefore, how to effectively model background to suppress background and highlight anomalies is important for HAD. Compared to background, anomalies can be characterized by low occurrence probability and high deviation from surrounding background distribution. Several existing HAD methods exploit these two properties to distinguish anomalies from complex background [7]–[16].

Recently, deep learning [17]–[20] has been successfully applied to HAD due to its powerful ability of feature learning. These deep learning-based methods can be categorized into reconstruction error-based methods [15], [16], [21]–[26] and feature extraction-based hybrid methods [27]–[29]. The former ones detect anomalies based on the assumption that reconstruction errors of anomalies are higher than those of background. The latter ones use deep neural networks as feature extractors and then detect anomalies in the deep feature space using conventional detectors. Although these methods have achieved promising results, there are several problems that need to be tackled.

1) Since feature learning and anomaly detection are separately performed, the extracted features are not really adapted to the subsequent anomaly detection task.
2) To enhance the discrimination between background and anomalies, existing methods [16], [22], [25] constrain the latent representations to obey a Gaussian distribution. However, it is hard to meet the Gaussian assumption in practice.
3) Although spatial information has been proven to be useful for HAD [30], it is rarely exploited for feature learning by existing methods, leading to suboptimal detection performance.

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4) The negative effect of redundant band is neglected, which may weaken the differences between background and anomalies and thus deteriorate the detection performance.

To address the aforementioned issues, we propose an end-to-end trainable deep one-class classification network, namely, spectral–spatial deep support vector data description (ss-DSVDD) for HAD. Inspired by [7], our motivation is that HAD can be converted as a one-class classification task, which is based on that background is widely distributed and dominant in HSI data. Hence, there exist significantly imbalanced class sizes between background and anomalies. Specifically, background samples are mapped into the latent space to obtain the deep features. Since prior knowledge of background is unknown, a density clustering-based background purification method is proposed to obtain pseudo-pure background samples. Then, the deep features are fed into the deep one-class classifier to train a compact hypersphere. At the test phase, if the feature of a sample is outside the hypersphere, the sample is classified as an anomaly. In this way, feature learning and anomaly detection are integrated into an end-to-end trainable network. Meanwhile, the proposed method uses a one-class classifier to model the background without assuming its statistic distribution.

To make full use of both spectral information and spatial information, a two-stream deep convolutional autoencoder (DCAE) network is designed to extract spectral and spatial features, respectively. Inspired by [31], a spectral–spatial feature fusion strategy is proposed to fuse these two styles of features due to their complementarity. In this way, the fused features can better reconstruct the original spectral and spatial inputs during the training phase.

Moreover, since redundant bands may weaken the differences between background and anomalies, a channel attention-based band selection module is used to reduce the negative effect on detection performance, which is based on the raw spectral bands that can be reconstructed by an informative band subset [32]. The channel attention mechanism is used to automatically learn larger weights for informative bands to reconstruct the raw spectral bands.

The main contributions of this article are summarized as follows.

1) We propose an end-to-end trainable deep one-class classification network for HAD, where feature learning and anomaly detection are integrated into a united framework to improve the detection performance. To the best of our knowledge, it is the first attempt to introduce a deep one-class classification method for HAD.
2) We propose an unsupervised training samples construction method to select pseudo-pure background samples to train the deep one-class classification network.
3) We propose a two-stream DCAE network to extract the spectral and spatial features and fuse these two types of features by a novel feature fusion strategy due to their complementarity.

The rest of this article is organized as follows. Some related works are briefly reviewed in Section II. The proposed method is described in Section III. Section IV presents the experimental setup and results in detail. Section V concludes this article.

II. Related Works

In this section, we review major works on HAD and CAEs.

A. Hyperspectral Anomaly Detection

Reed–Xiaoli (RX) algorithm [33] is a classical statistic-based anomaly detector. It assumes that the background conforms to a multivariate Gaussian distribution and then detects the anomalies by calculating the Mahalanobis distance between each test pixel and the background. In implementation, global RX (GRX) estimates the background statistics (i.e., mean value and covariance matrix) on the whole HSI data, whereas local RX (LRX) uses a local dual window to estimate these two statistics [34]. However, the assumption that background obeys a Gaussian distribution rarely occurs in HSI data due to its complex spectral–spatial structure. Besides, the estimated background statistics are susceptible to contamination of anomalies, which can deteriorate detection performance [13]. Consequently, several algorithms have been proposed to solve the abovementioned problems. The kernel RX (KRX) [35] nonlinearly maps the HSI data into a high-dimensional feature space where the background can conform to a Gaussian distribution. However, background estimation in KRX is still sensitive to the contamination of anomalies. The robust nonlinear anomaly detection (RNAD) [36] performs a robust regression analysis in the kernel feature space to tackle this issue. Cluster-based anomaly detection (CBAD) [37] groups the HSI data into several clusters and then performs RX on each cluster to detect anomalies. In addition, many other algorithms are proposed to model the background without an explicit Gaussian distribution assumption. The support vector data description (SVDD) [7] detector and its kernel version [38] build a compact hypersphere with minimum volume to model the local background. The dual-window density (DWD)-based detector [8] calculates the local density of each test pixel to detect anomalies since anomalies are scarce with low density in HSI data. In [39], a vertex and edge-weighted graph is constructed to represent the intrinsic relationships among the pixels, where an abnormal pixel manifests distant relationships with its neighbors. The fractional Fourier entropy (FrFE) [40] enhances the discrimination between anomalies and background in the optimal fractional domain, including both spectrum and its Fourier transform information. The kernel isolation forest (iForest) [9] is based on that anomalies are more susceptible to isolation than the background in the kernel space.

The representation-based methods [5], [10], [11], [41]–[43] represent each pixel as a linear combination of the atoms from a background dictionary and assume that the background pixels can be well expressed, while the anomalies cannot or have large representation residuals. These methods are categorized into sparse representation (SR)-based methods [10], [11], [41], [42] and collaborative representation (CR)-based methods [5], [43] according to different regulation constraints

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of representation coefficients. SR-based methods employ a small subset of atoms to linearly represent each pixel under a sparsity-level constraint. In general, reconstruction errors [10], [11], [41] or the frequencies of the dictionary atoms for HSI reconstruction [42] are employed to measure the anomaly scores. CR-based methods use the ℓ_2-norm to constrain representation coefficients to encourage all atoms to participate in the representation process. The anomaly score of a pixel is measured by its representation residual.

The matrix decomposition-based methods [12]–[14], [44]–[51] assume that anomalies are characterized by sparsity, whereas background is characterized as low rank after removing the anomalies. According to the different matrix decomposition techniques, these methods are divided into robust principal component analysis (RPCA)-based [52] methods [47], [49], [51], go decomposition (GoDec)-based [53] methods [12], [13], and low-rank representation (LRR)-based [54] methods [14], [44]–[46], [48], [50]. The RPCA-based methods assume that the data lie in a single subspace and decompose the data into a low-rank part and a sparse part. However, the single subspace assumption is unpractical since there exist several materials in real HSI scenes [14], [48]. The GoDec-based methods decompose the data into a low-rank part, a sparse part, and a noise part by using the GoDec algorithm. In [12], the sparse part is directly used to calculate the anomaly value of each pixel. In [13], the low-rank part is used to estimate the background statistics, and then, a Mahalanobis distance-based anomaly detector is built to detect anomalies. The LRR-based methods represent each pixel as a linear combination of the bases in a dictionary and search the lowest rank representation for all pixels [14], [54]. The residual between the observed image and the recovered background is used to detect anomalies.

Recently, deep learning-based methods have achieved promising performance in HAD. According to whether the deep neural network is directly used to detect anomalies, these methods are divided into reconstruction error-based methods [15], [16], [21]–[26] and feature extraction-based hybrid methods [27]–[29], [55]. Since anomalies are rare in HSI data, they produce limited contributions to the model training [15]. Hence, the reconstruction errors of anomalies are higher than those of background in reconstruction error-based methods. To enhance the discrimination between background and anomalies, different constraints [16], [25], [26] are applied to the models. In [16], an encoder is constrained to produce latent variables that obey a unit Gaussian distribution. In [25], an adversarial latent discriminator is used to force the latent variables to obey a Gaussian distribution. Besides, a consistency-enhanced representation constraint is used to reconstruct background samples and a shrink constraint is used to penalize samples that deviate from the background distribution. In [26], an embedding manifold constraint is applied to the latent variables to preserve the intrinsic structure of HSI data. Background-anomaly separability with a generative adversarial network (BASGAN) [22] explicitly introduces the background-anomaly separability constraint to enhance the background reconstruction while avoiding the anomaly reconstruction. The feature learning-based hybrid methods detect anomalies in two stages: 1) employ deep models to extract spectral features and 2) perform conventional methods on the extracted features to detect anomalies. For instance, the low-rank embedded network (LREN) [28] learns spectral features and discriminative dictionary by jointly training the autoencoders and a Gaussian mixture model. Then, an LRR-based method is performed to detect anomalies in the deep latent space. The weakly supervised low-rank representation (WSLRR) method [56] estimates the background distribution in a weakly supervised manner. Then, an LRR-based method is proposed to exploit the background structures and reduce the reconstruction biases introduced by the previous background estimation.

Besides, several works [57], [58] propose weighting strategies to improve the detection performance of existing anomaly detection methods by assigning small weights on the background locations of the detection maps.

B. Convolutional Autoencoders

Let $\mathbf{X} = \{\mathbf{x}_i \in \mathbb{R}^b\}_{i=1}^N$ be an HSI sample set, where N and b denote the number of samples and spectral dimension, respectively. A CAE is composed of an encoder $f(\cdot; \mathbf{W}, \mathbf{b})$ and a decoder $g(\cdot; \tilde{\mathbf{W}}, \tilde{\mathbf{b}})$, where $\mathbf{W}$ and $\tilde{\mathbf{W}}$ denote the learnable weights and $\mathbf{b}$ and $\tilde{\mathbf{b}}$ denote the bias. The encoder extracts the latent representations of the HSI samples. Let $\mathbf{Z} = \{\mathbf{z}_i \in \mathbb{R}^l\}_{i=1}^N$ denote the output of the encoder, where l is the dimension of $\mathbf{z}_i$. $\mathbf{z}_i$ can be obtained according to

$$\mathbf{z}_i = f(\mathbf{x}_i; \mathbf{W}, \mathbf{b}) = \sigma(\mathbf{x}_i * \mathbf{W} + \mathbf{b}) \tag{1}$$

where $*$ is the convolution operator and σ is the activation function. Then, $\mathbf{Z}$ is fed into the decoder to reconstruct the input samples. Let $\hat{\mathbf{X}} = \{\hat{\mathbf{x}}_i \in \mathbb{R}^b\}_{i=1}^N$ denote the output of decoder. $\hat{\mathbf{x}}_i$ can be obtained according to

$$\hat{\mathbf{x}}_i = g(\mathbf{z}_i; \tilde{\mathbf{W}}, \tilde{\mathbf{b}}) = \sigma(\mathbf{z}_i * \tilde{\mathbf{W}} + \tilde{\mathbf{b}}). \tag{2}$$

To ensure that the reconstructed samples are approximate to the raw samples, the loss function of the autoencoder is defined as

$$\mathcal{L} = \frac{1}{2}\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2 \tag{3}$$

where $\|\cdot\|_F$ denotes the Frobenius norm.

III. Proposed Methodology

In this section, an ss-DSVDD method is proposed for HAD. The flowchart of the proposed method at the training stage is represented in Fig. 1. We briefly introduce each module of the proposed network and illustrate the implementation details.

A. Background Purification Module

Since the proposed method aims to learn a compact hypersphere to involve the background sample features, the key step is to obtain background samples. There are three requirements for obtaining background samples.

1) It is an unsupervised task since the prior knowledge about the targets and background is unknown.
2) The background samples need to include all background classes. If there are missing background

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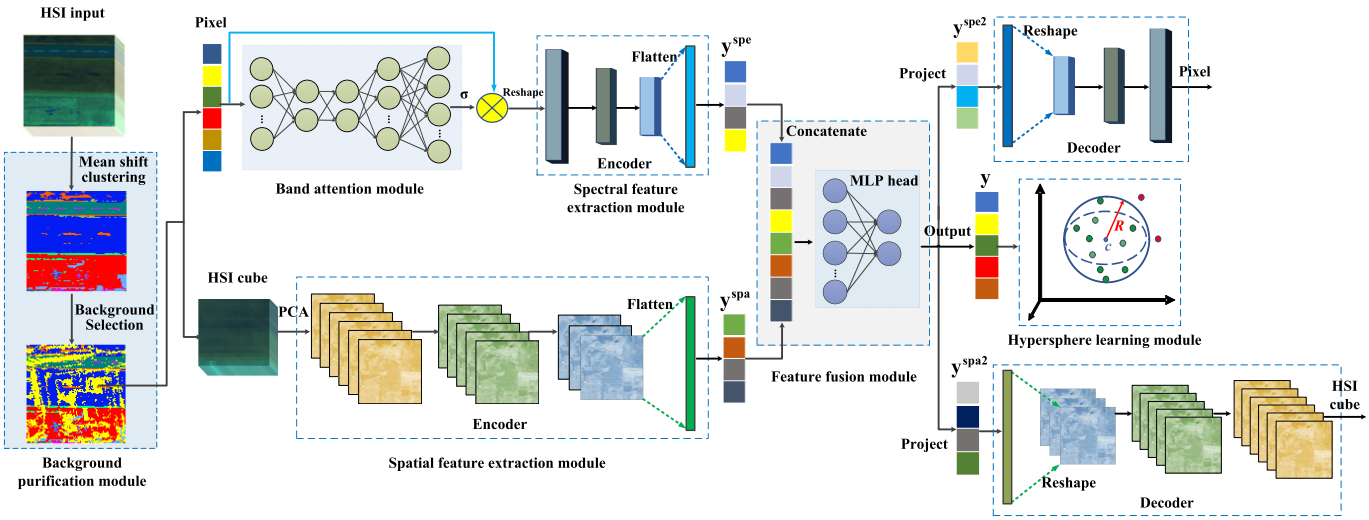


Fig. 1. Flowchart of our method at the training stage. Our method aims to train a deep one-class classifier for anomaly detection, which consists of six modules. (a) BPM is used to obtain the pseudo-pure background samples in yellow. (b) BAM is used to select informative bands to reconstruct the spectral signatures. (c) Spectral feature extraction module is based on deep 1-D convolutional autoencoders (CAEs). (d) Spatial feature extraction module is based on deep 2-D CAEs. (e) Feature fusion module fuses the spectral features and spatial features to generate joint spectral–spatial features, which are used to reconstruct the raw inputs and learn the hypersphere. (f) Hypersphere learning module constructs a hypersphere with minimum volume to involve the background features. (Best viewed in color.)

classes in the background samples, false alarms will be generated since samples of missing background classes will be identified as anomalies at the test stage.

3) The background samples should not involve abnormal samples. If the background is contaminated with anomalies, the learned hypersphere can represent anomaly features. Hence, similar abnormal samples will be identified as background at the test stage, resulting in a high missing alarm rate.

To meet the aforementioned requirements, a density clustering-based method is proposed to select pseudo-pure background samples in an unsupervised manner. The hyperspectral imagery (HSI) samples are first grouped into several clusters $[C_1, \ldots, C_n]$. Since mean-shift clustering [59] is a nonparameter kernel density estimation method, compared to the centroid-based clustering methods such as k-means, it can automatically partition the HSI samples without explicitly indicating the number of classes and is insensitive to class shapes. Therefore, the mean-shift clustering method is used to obtain a set of clusters. This method iteratively moves each sample toward the direction of local density gradient estimated by kernel function until convergence. The parameter bandwidth in this method denotes the window size that can determine the number of clusters. Generally, a large value of bandwidth will generate fewer clusters, while a small value will produce more clusters. Hence, a suitable value of bandwidth can facilitate the clustering performance. In this article, an effective method is adopted to automatically estimate the bandwidth bw by calculating the mean value of maximum distances between each sample and its k-nearest neighbors according to

$$bw = \frac{1}{N}\sum_{i=1}^{N} \max\{\|\mathbf{x}_i - \mathbf{x}_j\|^2, \mathbf{x}_j \in \mathcal{N}_k(\mathbf{x}_i)\} \tag{4}$$

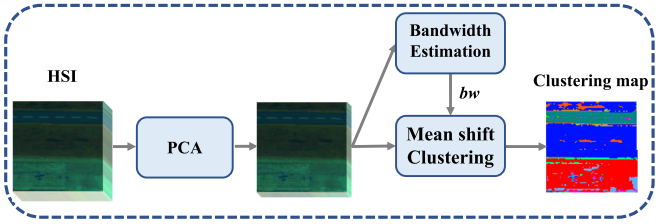


Fig. 2. Flowchart of the cluster generation. (Best viewed in color.)

where $\mathcal{N}_k(\mathbf{x}_i)$ denotes the set of neighbors of the sample $\mathbf{x}_i$. The number of nearest neighbors of each sample is calculated by $k = \tau \times N$, where τ is a quantile in (0,1]. Readers can refer to the sklearn library[1] for the implementation details. In summary, the flowchart of cluster generation is shown in Fig. 2. First, the HSI data are preprocessed by a principal component analysis (PCA) before the clustering process to reduce the computational cost. Second, the parameter bandwidth is estimated according to (4) to determine the number of clusters. Third, the clusters are obtained by performing mean-shift clustering on the HSI data.

Since anomalies have a small-size characteristic, once the clusters are generated, the isolated points or the sparse clusters with the number of samples lower than $\rho \times N$ are eliminated, where ρ is a parameter in (0,1]. However, there exist some samples at the edge of the remaining clusters (i.e., mixed pixels) that may be potential abnormal samples. To remove these samples, the Mahalanobis distance between each sample $\mathbf{x}$ and its corresponding cluster C_i is calculated as follows:

$$\mathrm{D}(\mathbf{x}) = (\mathbf{x} - \mu_i)^T \Gamma_i^{-1} (\mathbf{x} - \mu_i) \tag{5}$$

where μ_i and Γ are the mean value and the covariance matrix of the cluster C_i, respectively. The samples with higher

[1]https://scikit-learn.org/stable/modules/clustering.html#mean-shift

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Mahalanobis distances can be viewed as potential abnormal samples in each cluster. Then, the Mahalanobis distances of all samples in cluster C_i are sorted in ascending order. Finally, the samples located at the first 35% distances in each cluster are empirically selected as the pseudo-pure background samples. The implementation of background purification is illustrated in Algorithm 1.

Algorithm 1 Background Purification Algorithm

Input: HSI data X, ρ, τ, $S = \varnothing$
The bandwidth is estimated by Eq. (4).
Perform mean-shift algorithm on X to obtain clusters $[C_1, \cdots, C_n]$.
for $i = 1, \ldots, n$ **do**
  If $size(C_i) \leq \rho \times N$ **do**
    Eliminate the cluster C_i.
  end If
end for
Denote K as the number of remaining clusters.
for $i = 1, \ldots, K$ **do**
  for $j = 1, \cdots, size(C_i)$ **do**
    Calculate Mahalanobis distance d_j by Eq. (5).
  end for
  Sort the distance list $dist = [d_1, \cdots, d_j, \cdots]$ in ascending order.
  If $j \leq 0.35 \times len(dist)$ **do**
    Sample j is selected as background sample.
    $S \leftarrow S \cup \{j\}$.
  end if
end for
Output: Indexes of pseudo-pure background samples S.

B. Two-Stream Deep Autoencoders for Feature Extraction

The foundational component of the proposed method is the feature extraction module, which nonlinearly maps the raw HSI samples into a latent space. To exploit the spectral–spatial information, a two-stream DCAE architecture is designed as the backbone network.

1) Band Attention Module: Since redundant bands can weaken the differences between background and targets, it is expected to select a few informative bands to represent the complete spectral bands. Inspired by [32], the band selection can be converted into a band reconstruction task by using attention mechanism [60]. This mechanism selectively allocates available computational resources to the most informative parts of an input signal [61]. In this article, a band attention module (BAM) is added before the feature extraction to learn the band weights, which tends to assign larger weights to informative bands and hence reduces the negative effect of redundant bands. Let $\mathbf{X} = \{\mathbf{x}_i \in \mathbb{R}^b\}_{i=1}^m$ denote the pseudo-pure background samples, where m is the number of selected samples. As shown in Fig. 1, the BAM consists of five fully connected (FC) layers with 128-64-64-128-b neurons, and takes $\mathbf{X}$ as its input. It generates nonnegative band weights $\mathbf{U} = \{\mathbf{u}_i \in \mathbb{R}^b\}_{i=1}^m$. $\mathbf{u}_i$ is obtained according to

$$\mathbf{u}_i = \sigma(f(\mathbf{x}_i; \Theta_b)) \tag{6}$$

where $f(\cdot; \Theta_b)$ denotes the BAM with trainable parameters Θ_b and σ denotes the sigmoid activation function that is used to guarantee the nonnegativity of the learned weights. Subsequently, a bandwise multiplication operation is performed to obtain the reweighted bands that are formulated as follows:

$$\mathbf{H} = \mathbf{X} \otimes \mathbf{U} \tag{7}$$

where $\otimes$ denotes the bandwise multiplication and $\mathbf{H} = \{\mathbf{h}_i \in \mathbb{R}^b\}_{i=1}^m$ denotes the reweighted bands. Then, $\mathbf{H}$ is used to reconstruct $\mathbf{X}$ by the autoencoders. If $\mathbf{X}$ can be well reconstructed from $\mathbf{H}$, the band weights will reflect whether the corresponding bands are informative since these weights measure the band contributions to the spectral reconstruction.

2) Spectral Feature Extraction Module: This module is used to extract spectral features by using 1-D CAEs. Since each HSI pixel can be illustrated as a continuous spectral curve, the hierarchical architecture of 1D-CAEs is adopted to learn spectral features. This convolutional architecture can generate more discriminative features than the FC architecture. To this end, the reweighted tensor $\mathbf{H}^{m\times b}$ produced from the BAM is reshaped to $\mathbf{H}^{m\times b\times 1}$ to feed into the 1-D CAEs. Then, several convolutional layers are stacked to extract hierarchical features from $\mathbf{H}^{m\times b\times 1}$. Subsequently, the output of the last convolutional layer is flattened to form a 1-D vector. Finally, an FC layer is applied to this vector to obtain the latent representation of an HSI sample. Let $\mathbf{Y}_{\text{spe}} = \{\mathbf{y}_i^{\text{spe}} \in \mathbb{R}^p\}_{i=1}^m$ denote the spectral latent representations, where p is the dimension of the latent space. $\mathbf{y}_i^{\text{spe}}$ is obtained as follows:

$$\mathbf{y}_i^{\text{spe}} = f(\mathbf{h}_i; \Theta_{\text{spe}}) \tag{8}$$

where $f(\cdot; \Theta_{\text{spe}})$ denotes the encoder with parameters Θ_{spe}. Let $\hat{\mathbf{H}}^{m\times b\times 1}$ denote the output of the decoder. $\hat{\mathbf{H}}^{m\times b\times 1}$ is reshaped as $\hat{\mathbf{X}}^{m\times b}$ to calculate the reconstruction residual. The object function of the spectral reconstruction is defined as follows:

$$\mathcal{L}_{\text{spe}} = \frac{1}{2}\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2. \tag{9}$$

Once the network training is accomplished, the band weights can indicate the informative bands for the input samples. The bands with larger weights are considered as informative bands.

3) Spatial Feature Extraction Module: This module is used to extract spatial features by using 2-D convolution autoencoders (2-D CAEs). To reduce the computational cost, PCA is performed on the HSI data to reduce spectral bands to 4, which can preserve at least 95% of the cumulative percentage of variance. Then, the image patches with a window size of $w \times w$ centered at each pseudo-pure background pixel are extracted as training samples to feed into the 2-D CAEs. The impact of window size on the detection performance is analyzed in Section IV-F5. Let $\mathbf{X}_{\text{train}} = \{\mathbf{x}_i \in \mathbb{R}^{w\times w\times 4}\}_{i=1}^m$ denote the image patches. Several convolutional layers are stacked to extract hierarchical features from $\mathbf{X}_{\text{train}}$. Then, the output of the last convolutional layer is flattened to form a 1-D vector. Afterward, an FC layer is applied to this vector to obtain the spatial latent representation of an HSI patch. Let $\mathbf{Y}_{\text{spa}} = \{\mathbf{y}_i^{\text{spa}} \in \mathbb{R}^q\}_{i=1}^m$ denote the spatial latent representations,

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where q is the dimension of the latent space. $\mathbf{y}_i^{\text{spa}}$ is obtained according to

$$\mathbf{y}_i^{\text{spa}} = f(\mathbf{x}_i; \Theta_{\text{spa}}) \tag{10}$$

where $f(\cdot; \Theta_{\text{spa}})$ denotes the nonlinear encoder with parameters Θ_{spa}. The loss function of the autoencoders is defined as follows:

$$\mathcal{L}_{\text{spa}} = \frac{1}{2}\|\mathbf{X}_{\text{train}} - \hat{\mathbf{X}}_{\text{train}}\|_F^2 \tag{11}$$

where $\hat{\mathbf{X}}_{\text{train}}$ is the reconstructed image patches.

C. Feature Fusion Module

Once both spectral features and spatial features are obtained for each sample, the feature fusion module is used to fuse these two types of features to form joint spectral–spatial features. To this end, a novel feature fusion mode is proposed to exploit the complementary information of the spectral and spatial features to improve learning performance. Specifically, the spectral features and spatial features are concatenated along the feature dimension to form new spectral–spatial features denoted as $\mathbf{Y}_{\text{cat}} = \{\mathbf{y}_i^{\text{cat}} \in \mathbb{R}^{p+q}\}_{i=1}^m$. It is obtained according to

$$\mathbf{Y}_{\text{cat}} = \text{Cat}(\mathbf{Y}_{\text{spe}}, \mathbf{Y}_{\text{spa}}) \tag{12}$$

where $\text{Cat}(\cdot)$ denotes the concatenation operation. To further enhance the feature representation ability, $\mathbf{Y}_{\text{cat}}$ is fed into a multilayer perceptron (MLP) head with 1024–512 neurons to learn the joint spectral–spatial features denoted as $\mathbf{Y} = \{\mathbf{y}_i \in \mathbb{R}^{512}\}_{i=1}^m$. $\mathbf{y}_i$ is obtained according to

$$\mathbf{y}_i = \mathbf{W}_2\big(\sigma\left(\mathbf{W}_1 \cdot \text{BN}(\mathbf{y}_i^{\text{cat}}) + \mathbf{b}_1\right)\big) + \mathbf{b}_2 \tag{13}$$

where $\text{BN}(\cdot)$ is the batch normalization operation [62] and σ is a rectified linear unit (ReLU) [63] activation function. $\mathbf{W}_1 \in \mathbb{R}^{1024\times(p+q)}$, $\mathbf{b}_1 \in \mathbb{R}^{1024}$, $\mathbf{W}_2 \in \mathbb{R}^{512\times1024}$, and $\mathbf{b}_2 \in \mathbb{R}^{512}$ are learnable weights and biases of the MLP head. Then, $\mathbf{Y}$ is used to build the one-class classifier and reconstruct the raw spectral pixels and image patches. To implement the reconstruction, $\mathbf{Y}$ is nonlinearly projected to original spectral and spatial latent space to preserve consistent dimensions with $\mathbf{Y}_{\text{spe}}$ and $\mathbf{Y}_{\text{spa}}$. Let $\mathbf{Y}_{\text{spe2}} = \{\mathbf{y}_i^{\text{spe2}} \in \mathbb{R}^p\}_{i=1}^m$ and $\mathbf{Y}_{\text{spa2}} = \{\mathbf{y}_i^{\text{spa2}} \in \mathbb{R}^q\}_{i=1}^m$ denote the projection spectral features and spatial features, respectively. $\mathbf{y}_i^{\text{spe2}}$ and $\mathbf{y}_i^{\text{spa2}}$ can be obtained by separately using an FC layer as follows:

$$\mathbf{y}_i^{\text{spe2}} = \sigma(\mathbf{W}_3 \cdot \mathbf{y}_i + \mathbf{b}_3) \tag{14}$$

$$\mathbf{y}_i^{\text{spa2}} = \sigma(\mathbf{W}_4 \cdot \mathbf{y}_i + \mathbf{b}_4) \tag{15}$$

where σ is the ReLU activation function and $\mathbf{W}_3 \in \mathbb{R}^{p\times512}$, $\mathbf{b}_3 \in \mathbb{R}^p$, $\mathbf{W}_4 \in \mathbb{R}^{q\times512}$, and $\mathbf{b}_4 \in \mathbb{R}^q$ are learnable weights and biases of these two FC layers. In this way, $\mathbf{Y}_{\text{spe2}} \in \mathbb{R}^{m\times p}$ and $\mathbf{Y}_{\text{spa2}} \in \mathbb{R}^{m\times q}$ can be used to reconstruct the raw spectral signatures and image patches by using decoders. Moreover, since $\mathbf{Y}_{\text{spe2}}$ and $\mathbf{Y}_{\text{spa2}}$ contain both spectral information and spatial information, they can better reconstruct the raw inputs. Finally, $\mathbf{Y}_{\text{spe2}}$ and $\mathbf{Y}_{\text{spa2}}$ are separately fed into an FC layer and several convolutional transpose layers to generate the reconstructed pixels and image patches, respectively.

TABLE I
NETWORK SETTINGS OF THE TWO-STREAM CAES

CAEs	layers	kernel size	channels	BN	bias	stride	ReLU
1-D CAEs	encoder1	11×1	120	Yes	no	2	Yes
	encoder2	7×1	80	Yes	no	2	Yes
	encoder3	5×1	50	No	no	2	Yes
	decoder1	5×1	50	Yes	no	2	Yes
	decoder2	7×1	80	Yes	no	2	Yes
	decoder3	11×1	120	No	no	2	Yes
2-D CAEs	encoder1	3×3	128	Yes	no	1	Yes
	encoder2	3×3	64	Yes	no	1	Yes
	encoder3	3×3	32	No	no	1	Yes
	decoder1	3×3	32	Yes	no	1	Yes
	decoder2	3×3	64	Yes	no	1	Yes
	decoder3	3×3	128	No	no	1	Yes

D. Hypersphere Learning Module

Inspired by the successful application of SVDD [7] in HAD, a hypersphere learning module extends SVDD to deep feature space to improve its detection performance. The deep support vector data description (DSVDD) classifier is trained to involve the background features into a hypersphere with minimum volume. A test sample is classified as an anomaly when it is outside the hypersphere. The loss of the DSVDD classifier is defined as

$$\mathcal{L}_{\text{DSVDD}} = \min_R R^2 + \frac{1}{\nu m}\sum_{i=1}^m \max\{0, \|\mathbf{y}_i - \mathbf{c}\|^2 - R^2\} \tag{16}$$

where $\mathbf{y}_i \in \mathbf{Y}^{m\times512}$ denotes the joint spectral–spatial feature of a background sample, $\mathbf{c} \in \mathbb{R}^{512}$ denotes the center of the hypersphere, $R \in \mathbb{R}^{512}$ denotes the trainable radius of the hypersphere, and $\nu \in (0, 1]$ is a hyperparameter that controls the tradeoff between the volume of the hypersphere and the number of feature points that outside the hypersphere. The first term is used to minimize the volume of the hypersphere. The second term is used to penalize the points whose distances to the center are larger than the radius. Note that, since the proposed network models the background by a compact hypersphere, there is no explicit statistic distribution assumption on the background.

E. Implementation Details

To preserve the local structure of HSI data, the spectral reconstruction and spatial reconstruction terms are jointly optimized with the DSVDD term during the training stage. In this way, feature learning and anomaly detection are integrated into a united object function with definite physical meaning. The object function of the proposed ss-DSVDD network is defined as

$$\mathcal{L} = \mathcal{L}_{\text{DSVDD}} + \lambda_1\mathcal{L}_{\text{spe}} + \lambda_2\mathcal{L}_{\text{spa}} \tag{17}$$

where λ_1 and λ_2 are the weights to balance the contributions of different terms. Since the size of training samples for HAD is limited (e.g., in the order of thousands of samples), the proposed network should remain a lightweight size. Therefore, the two-stream autoencoder network is designed with shallow layers, whose channel numbers and convolutional kernel sizes are reported in Table I. Moreover, to avoid directly training the

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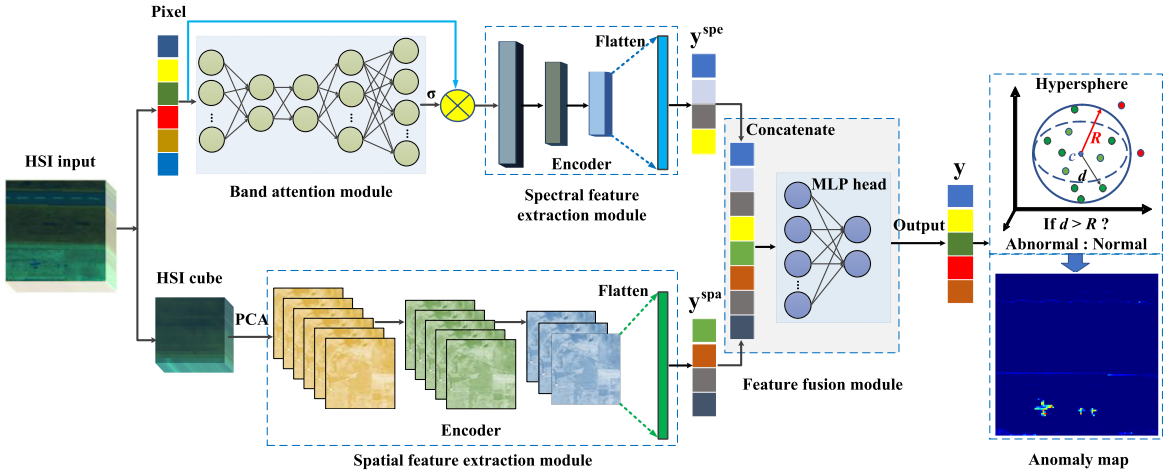


Fig. 3. Flowchart of the proposed method at the test stage. First, the spectral and spatial features of each test sample are extracted by the two-stream encoders. Second, they are concatenated and fed into the feature fusion module to generate joint spectral–spatial features. Third, if the distance d from the joint spectral–spatial feature to the center of the hypersphere c is larger than the radius of the hypersphere R, the corresponding test sample is classified as an anomaly one. (Best viewed in color.)

network from scratch, a two-stage pipeline strategy is adopted to train the proposed network:

1) *Pretraining Stage:* Pretraining aims to obtain good initialization weights and reduce the reconstruction difficulty at the subsequent fine-tuning stage. At the pretraining stage, the background samples are selected first. Then, the reconstruction losses $\mathcal{L}_{\text{spe}}$ and $\mathcal{L}_{\text{spa}}$ are used to update the two-stream DCAEs. After the pretraining stage, the center $\mathbf{c}$ of the hypersphere is empirically set as the mean of joint spectral–spatial features, which are obtained by performing an initial forward pass on the selected background samples.
2) *Fine-Tuning Stage:* At this stage, we fine-tune our network using the pretrained model for initialization. First, the radius of the hypersphere is initially set to 0.01. Second, the proposed network is trained with the loss function in (17). After T_0 epochs, the radius R is updated by the $(1-\nu)$-quantile of the distances from the joint spectral–spatial features of a minibatch of samples to $\mathbf{c}$. When the iteration number reaches the maximum iteration $T_{\max}$, the training is stopped.

The training process of our network is illustrated in Algorithm 2.

At the test stage, the anomaly score of a test sample $\mathbf{x}_i \in \mathbf{X}^{N\times b}$ is defined as the distance between its corresponding deep feature and the center of the hypersphere, i.e.,

$$s(\mathbf{x}_i) = \|\phi(\mathbf{x}_i;\Theta) - \mathbf{c}\|^2 - R^2 \tag{18}$$

where $s(\cdot)$ denotes the anomaly score of a test sample and $\phi(\cdot;\Theta)$ denotes the proposed network with parameters Θ. The flowchart at the test phase is presented in Fig. 3. The decoders are removed from the trained network at the test stage. The detection scores are obtained by performing a forward pass on the test samples with (18). If the anomaly score of a test sample is positive, the sample is classified as an anomaly. Otherwise, the sample is classified as background.

Algorithm 2 Training Process of the ss-DSVDD Network

Input: $\mathbf{X}$, parameters p, q, w, ν, λ_1, λ_2, maximum iteration T_{max}, T_0, and $t = 1$.

1: Select training samples by algorithm 1.
2: Pre-train the two-stream deep convolutional autoencoders by Eq. (9) and Eq. (11).
3: Calculate the center of hypersphere $\mathbf{c}$ as the mean of joint spectral-spatial features.
4: Initialize the radius R of the hypersphere as 0.01.
5: **while** $t \leq T_{max}$ **do**
6: Train the whole network by Eq. (17).
7: **if** $t \geq T_0$ **do**
8: Update the radius R.
9: **end if**
10: $t \leftarrow t + 1$.
11: **end while**.

Output: Network parameters, center $\mathbf{c}$, and radius R.

IV. Experimental Results and Discussion

In this section, we conduct extensive experiments to evaluate the detection performance of the proposed method on five widely used hyperspectral datasets.

A. Datasets Description

The detection performance of the proposed method is evaluated on five publicly available hyperspectral datasets, whose details are shown in Table II. These hyperspectral datasets are briefly introduced as follows.

1) HYDICE Urban Dataset: This dataset was captured by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) airborne sensor over an urban area, CA, USA. It has 162 spectral bands ranging from 400 to 2500 nm. The spatial resolution is 1 m/pixel. The original image has a spatial size of 307 × 307, while a subimage with a size of 80 × 100 is selected in this article. For each pixel, 162 bands are preserved to retain useful information. There are 17 abnormal pixels that need to be detected, accounting for 0.21% of the total

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TABLE II
DETAILS OF THE HYDICE URBAN, CAT ISLAND, GULFPORT, SANDIEGO, AND EI SEGUNDO DATASETS

Dataset	sensors	spatial resolution	wavelength range	image size	anomaly size	anomaly proportion
HYDICE Urban	HYDICE	1 m/pixel	400-2500 nm	$80 \times 100 \times 162$	17	0.21%
Cat Island	AVIRIS	17.2 m/pixel	unknown	$150 \times 150 \times 188$	19	0.08%
Gulfport	AVIRIS	3.4 m/pixel	400-2500 nm	$100 \times 100 \times 191$	60	0.6%
San Diego	AVIRIS	3.5 m/pixel	370-2510 nm	$100 \times 100 \times 189$	134	1.34%
EI Segundo	AVIRIS	7.1 m/pixel	366-2496 nm	$250 \times 300 \times 224$	2048	2.73%

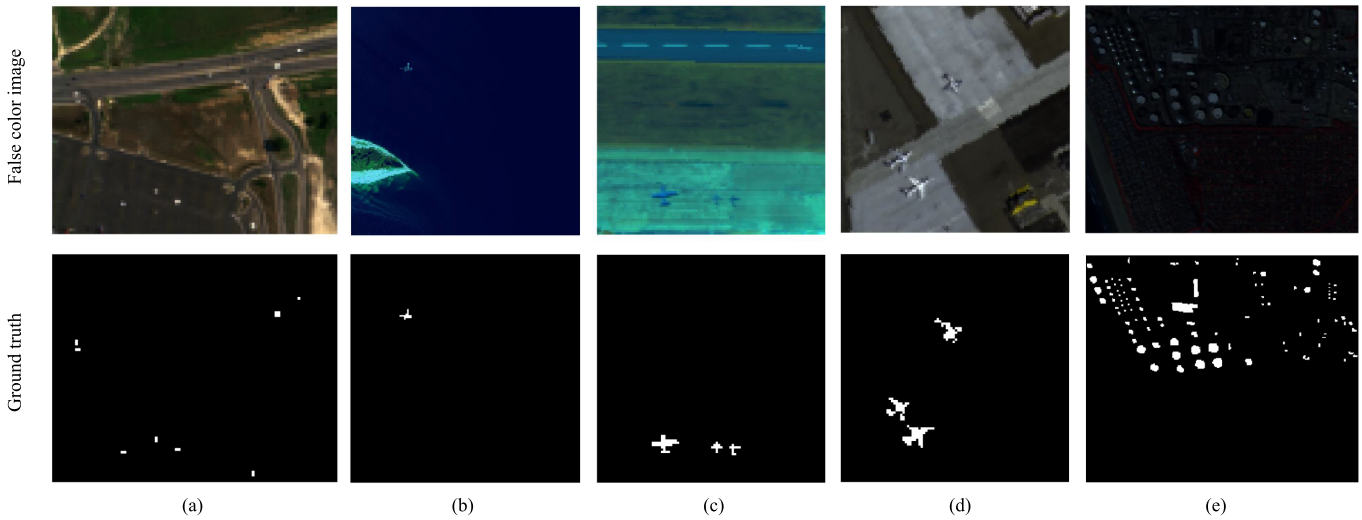


Fig. 4. Datasets used in the experiments. The first row shows the false-color images, and the second row shows the corresponding ground-truth maps of the datasets. (a) HYDICE Urban. (b) Cat Island. (c) Gulfport. (d) SanDiego. (e) EI Segundo.

number of pixels. These anomaly targets are cars and roofs. The false-color image of this dataset and its corresponding ground-truth map are shown in Fig. 4(a).

2) Cat Island Dataset: This dataset [30] was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor on September 12, 2010. The spatial resolution of this dataset is 17.2 m/pixel, and the spatial size is 150×150. After removing the unsuitable bands, 188 bands are retained. As shown in Fig. 4(b), there is an airport with 19 anomaly pixels to be detected in the dataset, accounting for 0.08% of the total number of pixels.

3) Gulfport Dataset: This dataset [30] was captured by the AVIRIS sensor from Gulfport, Southern Mississippi, USA, in 2010. The spatial resolution is 3.4 m/pixel. It has 191 spectral bands ranging from 400 to 2500 nm with a spatial size of 100×100. As shown in Fig. 4(c), there are three airplanes in various scales with 60 anomalous pixels to be detected in this dataset, accounting for 0.6% of the total number of pixels.

4) San Diego Dataset: This dataset was also captured by the AVIRIS sensor over the San Diego airport area, CA, USA. It has 224 spectral bands ranging from 370 to 2510 nm, among which 189 bands are preserved after eliminating several water absorption bands and low-SNR bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224). The spatial resolution is 3.5 m/pixel and the spectral resolution is 10 nm. The spatial size of each band is 100×100. As shown in Fig. 4(d), there are three airports with 134 anomaly pixels to be detected in the dataset, accounting for 1.34% of the total number of pixels.

5) EI Segundo Dataset: This dataset was also captured by the AVIRIS sensor over an area that covers an oil refinery, a few residential areas, parks, and a school zone. The spatial resolution is 7.1 m/pixel and the spatial size is 250×300. This dataset has 224 spectral bands ranging from 366 to 2496 nm. As shown in Fig. 4(e), there are 2048 anomaly pixels to be detected in the dataset, accounting for 2.73% of the total number of pixels.

B. Compared Methods and Evaluation Criteria

1) Compared Methods: We compare the proposed method with several existing hyperspectral anomaly detectors, including LRX [34], QLRX [64], SVDD [7], CRD [43], SR-LMM [10], LRASR [14], and DAEAD [15]. The LRX and QLRX are statistic-based methods. The CRD and SR-LMM are representation-based methods. The LRASR is a matrix decomposition-based method. DAEAD is a deep learning-based method. SVDD is also a one-class classifier for HAD. All parameters of the compared clustering algorithms are empirically tuned to their optimum.

The hyperparameters of the proposed method are reported in Table III. ν is set to 0.9 in all experiments. The Adam optimizer [65] is used with the learning rate being set to 10^{-4}. The maximum number of training epochs is $T_{\max} = 100$ and the radius R is updated after $T_0 = 10$ epochs.

2) Evaluation Criteria: Evaluation of detection performance on different detectors includes qualitative evaluation and quantitative evaluation. Color anomaly map,

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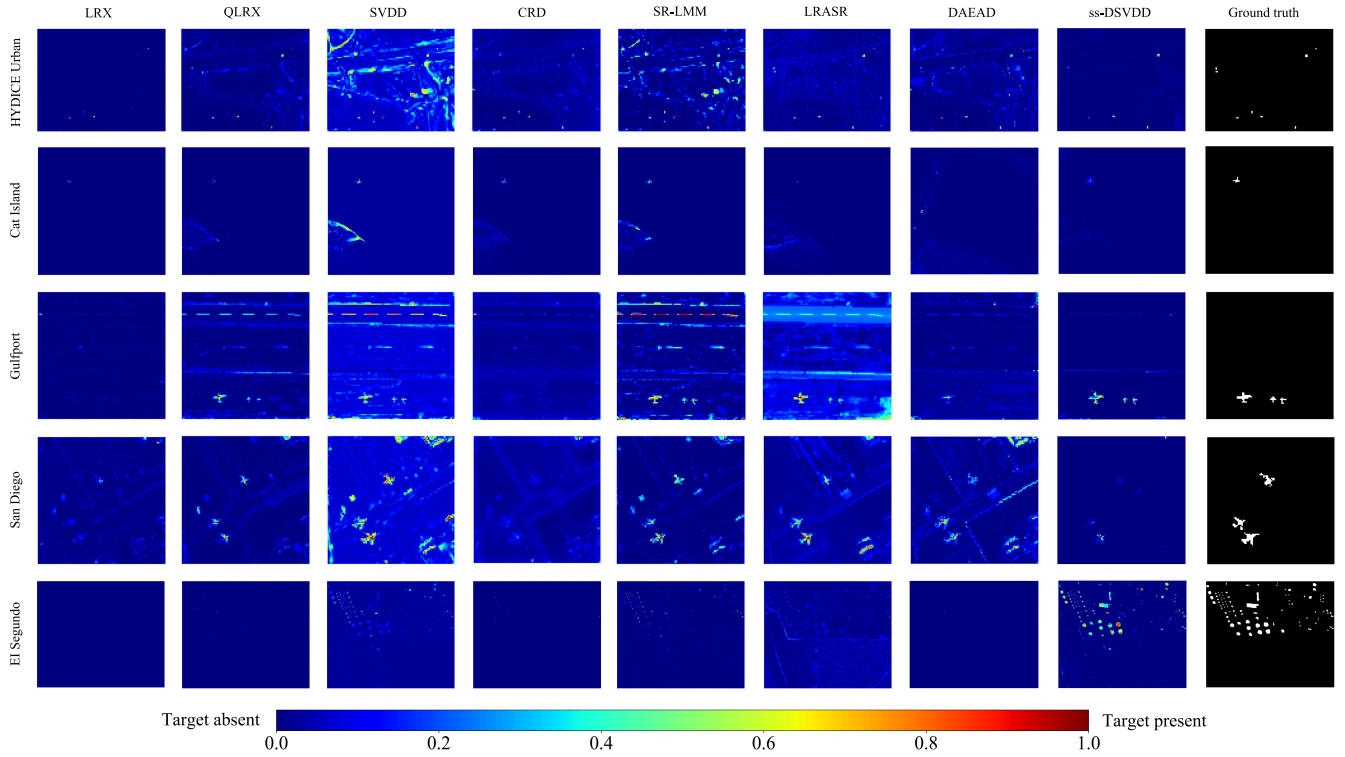


Fig. 5. Color anomaly maps obtained by different methods on five datasets. Last column: ground-truth maps.

background-anomaly separation map, and receiver operating characteristic (ROC) curve are used as qualitative evaluation metrics. Color anomaly map reflects the detection performance on both background suppression and target enhancement. Background-anomaly separation map reflects the distribution characteristic of background and anomalies of detection results. The ROC curve reflects the tradeoff between the probability of detection (PD) and false alarm rate (FAR). When a given threshold is applied to the detection output map, these two probabilities can be calculated as follows:

$$\mathrm{PD} = \frac{N_D}{N_T}, \quad \mathrm{FAR} = \frac{N_F}{N_B} \tag{19}$$

where N_T is the total number of true target pixels in the dataset, N_D is the number of true target pixels that are detected as targets, N_B is the total number of background pixels in the dataset, and N_F is the number of background pixels that are determined as targets. ROC curve plots the PD versus FAR under a series of continuous thresholds. When an FAR is fixed, the detection algorithm with higher PD scores performs better than that with lower PD scores.

The area under curve (AUC) obtained by integrating the area under the ROC curve is used to quantitatively evaluate the detection performance. The ideal value of AUC is 1. In addition, the running time is reported to evaluate the efficiency of different methods.

C. Detection Results and Analyses

1) Qualitative Results: The color anomaly maps generated by different detection methods are visualized in Fig. 5. Although QLRX, SVDD, SR-LMM, and LRASR can highlight the anomaly targets in the scenes, these methods produce more false alarms than the proposed method, especially at the edges of different materials. On the contrary, LRX, CRD, and DAEAD can achieve lower responses at the background regions, but some anomaly targets are missing in their maps. For example, these methods cannot detect complete targets with different scales on the Gulfport dataset. In contrast, the proposed method generates satisfactory anomaly maps on both background suppression and anomaly target enhancement, which visually demonstrates the superior detection performance over other methods.

TABLE III
HYPERPARAMETER SETTINGS OF THE PROPOSED METHOD

dataset	τ	w	p	q	ρ	λ_1	λ_2
HYDICE Urban	0.075	9	32	7	0.005	10	0.01
Cat Island	0.3	9	64	512	0.005	0.05	5
Gulfport	0.1	9	32	7	0.003	10	0.0001
San Diego	0.075	5	32	7	0.01	10	0.05
EI Segundo	0.3	5	32	7	0.1	10	0.001

The normalized background-anomaly separation maps of different detectors are shown in Fig. 6. The yellow box denotes anomaly targets, and the blue box denotes background pixels. The interval between the red box and the blue box denotes the separability between anomalies and background. Moreover, the height of the blue box denotes the background suppression degree of these detectors. It can be observed that the heights of the blue boxes of the proposed method are relatively lower than those of other methods on the Cat Island, Gulfport, SanDiego, and EI Segundo datasets, which demonstrates that the proposed method achieves better

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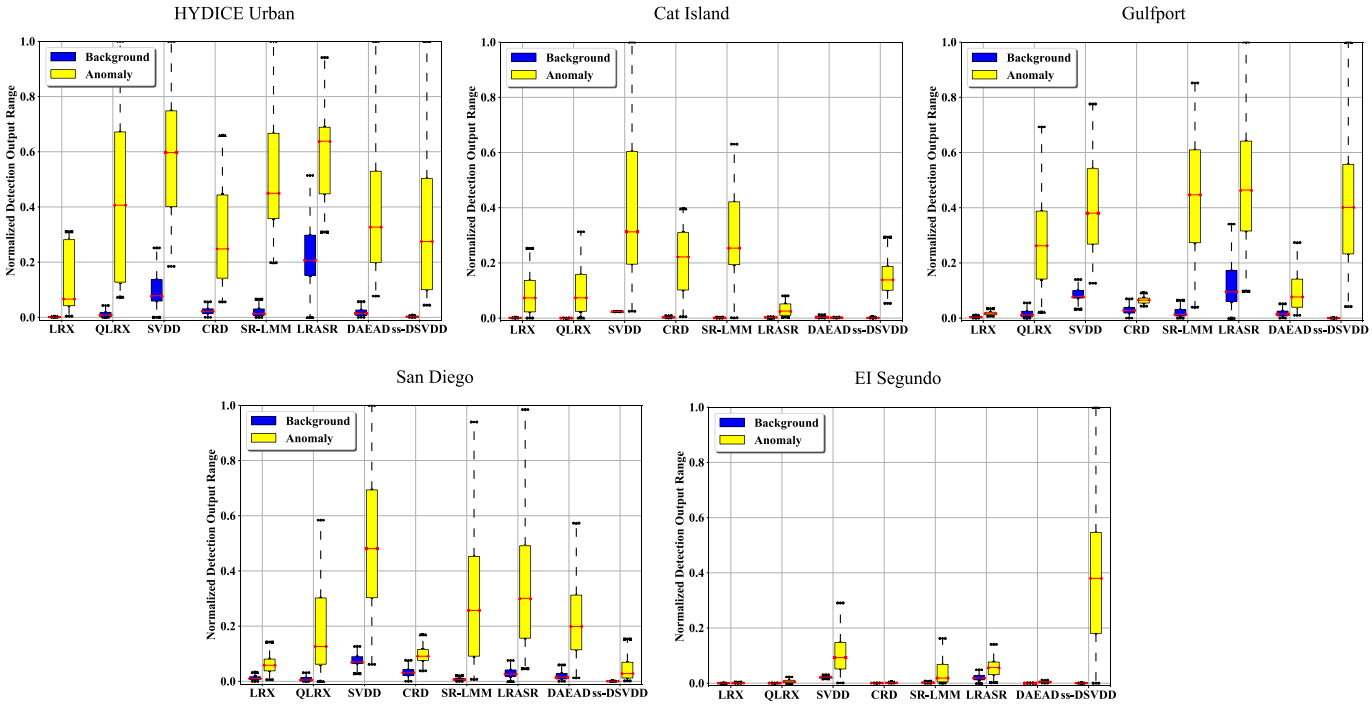


Fig. 6. Separation maps obtained by the compared methods on five datasets.

TABLE IV
AUC SCORES (%) OF DIFFERENT DETECTION METHODS ON FIVE DATASETS

Dataset	LRX	QLRX	SVDD	CRD	SR-LMM	LRASR	DAEAD	ss-DSVDD
HYDICE Urban	99.61	99.63	97.60	99.46	99.03	99.24	99.31	**99.99**
Cat Island	94.74	98.19	98.86	98.87	98.43	94.07	96.31	**99.98**
Gulfport	96.17	97.64	96.91	94.89	96.91	95.10	91.24	**99.92**
San Diego	93.81	95.83	96.88	95.89	96.71	98.39	96.10	**99.53**
EI Segundo	92.33	96.22	94.56	94.94	86.53	84.84	99.06	**99.49**

background suppression performance. Meanwhile, there are almost no overlaps between the blue boxes and yellow boxes of the proposed method on the aforementioned datasets, which shows better separability between anomalies and background than its counterparts. Although the blue box of our method is higher than that of the LRX method on the HYDICE Uban dataset, the interval between the red box and the blue box of the proposed method is larger than that of the LRX method on this dataset. Hence, the proposed method achieves better separability compared to the LRX method. Note that, SR-LMM achieves better separability than the proposed method on the HYDICE Uban dataset. However, its background suppression degree is weaker than that of the proposed method. Correspondingly, SR-LMM generates more false alarms on this dataset.

The ROC curves of different methods on the five datasets are shown in Fig. 7. It can be observed that the ROC curves of the proposed method are always above those of other methods on the HYDICE Urban and Gulfport datasets. Hence, the proposed method obtains the highest PD among these methods. For the remaining datasets, when the FAR is larger than 0.001, the proposed method achieves the best performance. Moreover, the ROC curves achieved by the proposed method are nearest the top-left corner, which shows its better detection accuracies than other detectors on all datasets. Correspondingly, the proposed method can obtain the best AUC scores.

2) Quantitative Results: The AUC scores achieved by different methods on the five datasets are presented in Table IV, where the best result of each row is highlighted in bold. The following observations can be drawn from these results.

1) The proposed method achieves the highest AUC scores on the five datasets and outperforms all the compared methods by a notable margin, which is consistent with the areas under the ROC curves given in Fig. 7.
2) Compared with the conventional SVDD method [7], the proposed method significantly improves the detection performance on all the datasets. Specifically, the AUC scores achieved by the proposed method are improved by 2.39%, 1.12%, 3.01%, 2.65%, and 4.93% on the HYDICE Urban, Cat Island, Gulfport, San Diego, and EI Segundo datasets, respectively. This is because the proposed method detects the anomalies in the deep feature space, where anomalies can be better identified than performing detection in the original space. Meanwhile, the proposed method exploits the spatial information of each sample to improve the detection performance.

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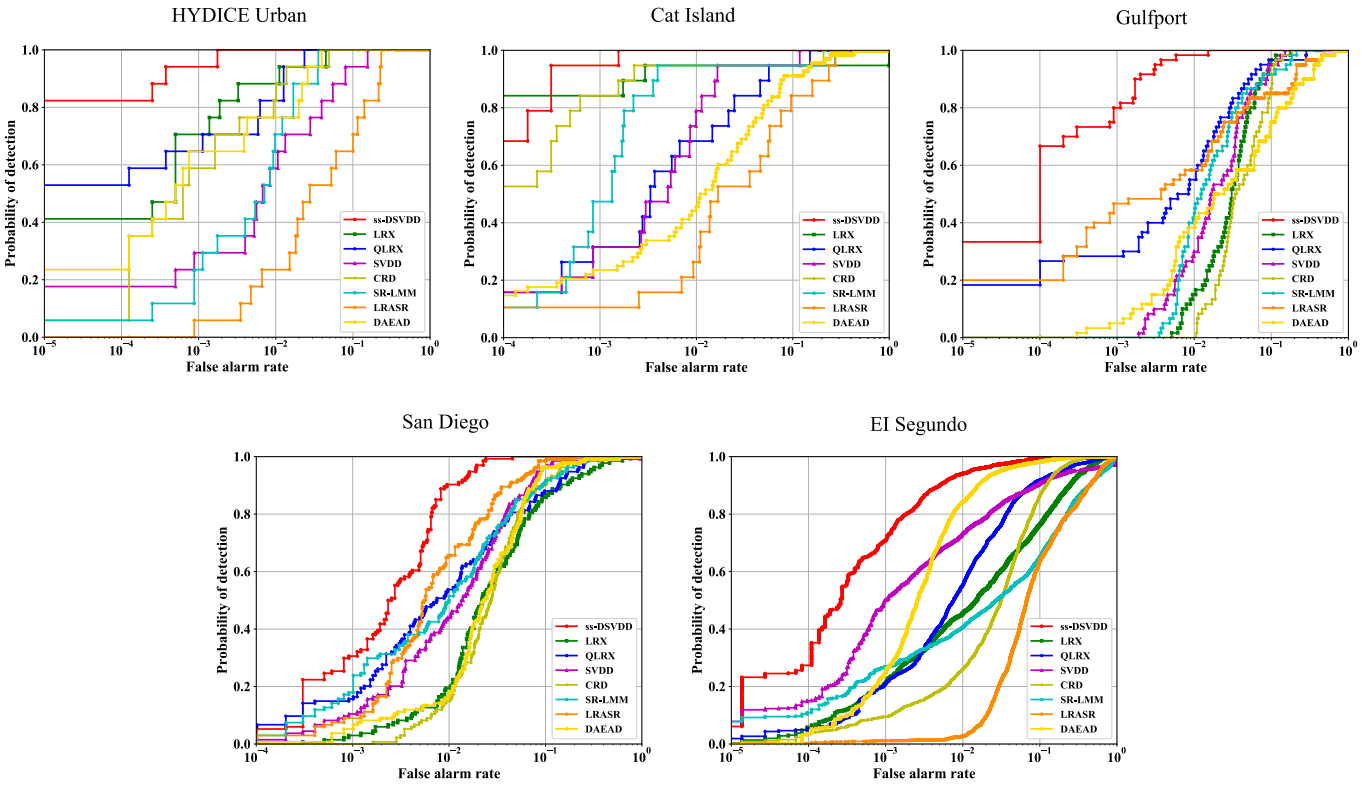


Fig. 7. ROC curves obtained by the compared methods on five datasets.

TABLE V
ABLATION STUDY OF THE PROPOSED METHOD ON FIVE DATASETS OVER AUC SCORES (%)

Component	HYDICE Urban	Cat Island	Gulfport	San Diego	EI Segundo
FC-DSVDD	89.47	92.32	86.36	80.06	58.68
FC-DSVDD with BPM	99.46	94.65	89.66	93.71	98.01
FC-DSVDD with BAM	90.60	92.86	87.20	87.67	60.49
FC-DSVDD with BPM and BAM	99.62	95.87	96.58	98.12	98.98
1-D DSVDD	99.00	96.66	97.93	96.21	96.55
1-D DSVDD with BPM	99.62	98.04	98.83	98.09	98.56
1-D DSVDD with BAM	99.59	97.76	98.16	96.56	97.14
1-D DSVDD with BPM and BAM	99.77	98.05	99.06	98.60	98.74
2-D DSVDD	86.43	98.19	97.01	94.55	96.57
2-D DSVDD with BPM	94.15	99.52	97.91	96.82	98.23
ss-DSVDD-FC	99.68	99.91	99.33	98.77	99.12
ss-DSVDD	99.99	99.98	99.92	99.53	99.49

3) The deep learning-based methods, such as DAEAD and ss-DSVDD, perform better than conventional methods on the EI Segundo dataset, where the anomaly targets are densely distributed. The conventional methods achieve lower AUC scores since the surrounding background of each test pixel is easily contaminated by the target samples. In contrast, the two deep learning-based methods achieve higher AUC scores on the dataset, which demonstrates their robustness to the dense target scene. That is because both DAEAD and ss-DSVDD are global detectors. Therefore, they are relatively insensitive to local background contamination.

D. Ablation Study

To analyze the effect of each module on the final detection results, an ablation study is conducted on all datasets over the AUC scores. We set three baseline methods by applying the DSVDD classifier to FC architecture autoencoders with 80-50-p-50-80 neurons, 1-D CAEs, and 2-D CAEs, respectively. For convenient description, they are denoted as FC-DSVDD, 1-D DSVDD, and 2-D DSVDD. Accordingly, the model that replaces the 1-D CAEs in the ss-DSVDD by FC autoencoders is denoted as ss-DSVDD-FC. Their object functions are similar to (16). The experimental results are reported in Table V. To validate the effectiveness of the background

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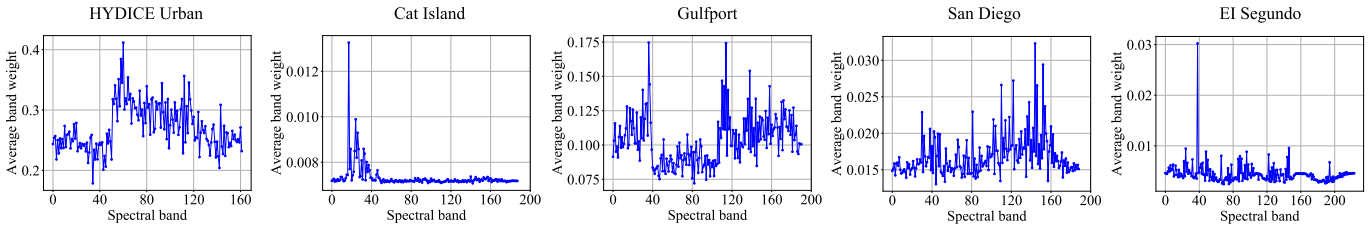


Fig. 8. Average band weights of the test samples on five datasets.

purification module (BPM), only the BPM is performed on the FC-DSVDD, 1-D DSVDD, and 2-D DSVDD methods. It can be observed that the AUC scores are significantly improved by using BPM on FC-DSVDD, 1-D DSVDD, and 2-D DSVDD. Moreover, when only the BAM is added to FC-DSVDD, the AUC scores are improved by 1.13%, 0.54%, 0.84%, 7.61%, and 1.81% on the five datasets. When only the BAM is added to 1-D DSVDD, the AUC scores are still improved by 0.59%, 1.10%, 0.23%, 0.35%, and 0.59% on the five datasets. This is because the BAM can effectively suppress the band redundancy and, thus, it improves the detection performance. The AUC scores can be further improved by applying both BPM and BAM to FC-DSVDD and 1-D DSVDD, respectively, which demonstrates the effectiveness of combining BPM with BAM. In contrast, ss-DSVDD achieves the highest AUC scores on the five datasets, which validates that the proposed feature fusion strategy can effectively combine spectral and spatial features. Moreover, ss-DSVDD-FC achieves higher AUC scores than the methods without adopting the fusion strategy, which can also validate the effectiveness of the fusion strategy. Note that the AUC scores achieved by the methods using the 1-D convolutional architecture to extract spectral features are higher than those achieved by using FC architecture. Generally, HSI has a high correlation between neighboring bands. The 1-D convolutional layer can extract the spectral features from the local neighboring bands by convolutional kernel learning, while the FC layer uses the whole bands to learn the features and thus can introduce irrelative band information. Therefore, the convolutional architecture can extract more discriminative spectral features than the FC architecture.

E. Visualization of Band Attention Weights

The band attention weights reflect the band contributions to the spectral reconstruction. Following [32], the learned band attention weights are averaged to indicate the informative bands, i.e., the bands with large average weights are considered as informative bands. As shown in Fig. 8, the average band weights of the test samples on the five datasets are different with respect to the bands. Intuitively, the sparse peaks on the curves indicate that the corresponding bands are informative, which validates that the BAM can generate larger weights for informative bands. These bands make more contributions to spectral reconstruction.

F. Parameter Sensitivity Analyses

In this section, we conduct experiments to investigate the impact of the important parameters on the detection performance.

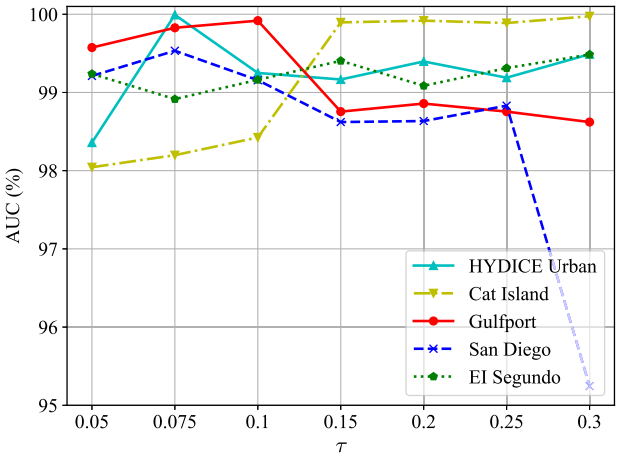


Fig. 9. AUC scores with respect to different settings of τ on five datasets.

1) Impact of τ: The quantile τ is the only parameter in the mean-shift clustering algorithm, which determines the number of clusters. Specifically, a small τ corresponds to a large number of clusters. We investigate the impact of τ on the detection performance and set $\tau \in \{0.05, 0.075, 0.1, 0.15, 0.2, 0.25, 0.3\}$ on the five real datasets. The influence of τ on the AUC scores is shown in Fig. 9. It can be seen that the AUC scores vary with respect to different values of τ on the five datasets. This is because the background structures are complex on the HYDICE Urban, Gulfport, and San Diego datasets. If the values of τ are large, the clusters segmented by the mean-shift clustering method are rare that may contain both background and anomaly targets. Hence, it is difficult to remove anomaly targets from the background. In contrast, since the background distributions of the Cat Island and EI Segundo datasets are relatively homogeneous, background and the target pixels can be effectively grouped into different clusters by using large values of τ. Correspondingly, the optimal values of τ are 0.3 on these two datasets.

2) Impact of ρ: Parameter ρ is used to determine whether a cluster is a potential anomaly cluster that contains rare pixels. The impact of ρ on detection performance is shown in Fig. 10, where we set $\rho \in \{0.001, 0.003, 0.005, 0.007, 0.01, 0.03, 0.05, 0.07, 0.1\}$. It can be clearly observed that the AUC scores achieved on the HYDICE Urban, Cat Island, and Gulfport datasets are higher than 99.00% with respect to different values of ρ. Hence, they are insensitive to ρ. Moreover, the AUC scores achieved on the San Diego dataset slightly fluctuate with respect to ρ. In contrast, the AUC scores achieved by the EI Segundo dataset are significantly influenced by ρ. Meanwhile,

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TABLE VI
RUNNING TIME OF DIFFERENT DETECTION METHODS ON FIVE DATASETS

Dataset	LRX	QLRX	SVDD	CRD	SR-LMM	LRASR	DAEAD	ss-DSVDD
HYDICE Urban (s)	35.76	1.95	7.95	8.76	5.06	26.18	41.30	107.39
Cat Island (s)	126.69	5.19	21.64	25.00	35.23	96.19	133.05	301.99
Gulfport (s)	71.06	2.98	13.31	17.53	26.30	31.68	53.78	131.54
San Diego (s)	67.89	2.89	12.34	13.55	25.23	28.70	47.93	160.35
EI Segundo (s)	456.68	21.51	95.23	89.33	101.55	414.85	374.61	645.74

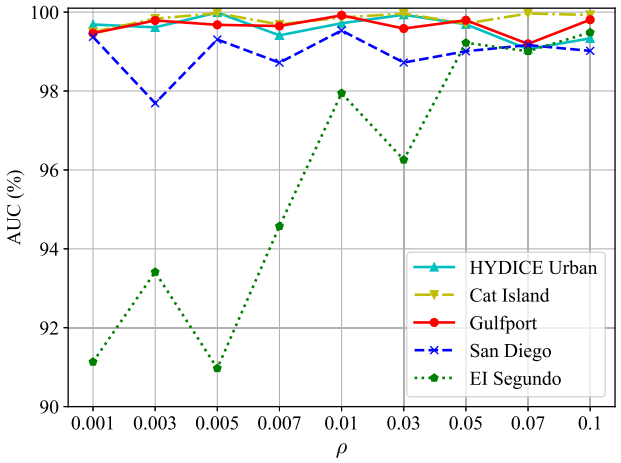


Fig. 10. AUC scores with respect to different settings of ρ on five datasets.

the AUC performance tends to increase as ρ increases. This is because anomaly targets are densely distributed on the EI Segundo dataset, and their number is large. Correspondingly, if ρ is too small, several clusters that contain anomaly targets cannot be eliminated from the background, resulting in missing detection at the test phase. In summary, the optimal ρ should be empirically set according to the target distribution on different datasets.

3) Impact of λ_1 and λ_2: Parameters λ_1 and λ_2 control the contributions of spectral reconstruction term and spatial reconstruction term in the total loss, respectively. The impact of λ_1 and λ_2 on the detection performance is presented in Fig. 11, where λ_1 and λ_2 are in the range of $\{0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05, 0.1, 5, 10\}$. It can be seen that the AUC scores achieved on the HYDICE Urban, San Diego, and EI Segundo datasets are relatively high when $\lambda_1 > 0.1$ and tend to decrease as λ_2 increases. Therefore, the spectral reconstruction term plays a more important role than the spatial reconstruction term in obtaining optimal detection results on these datasets. For the Cat Island dataset, when λ_2 is 5 or 10, the AUC scores are insensitive to λ_1. As λ_2 decreases, the AUC scores tend to decrease. Note that, most AUC scores are lower than 99.00% when $\lambda_1 \geq \lambda_2$. Therefore, the spatial reconstruction term contributes more than the spectral reconstruction term to the optimal detection results. The AUC scores achieved on the Gulfport dataset are insensitive to λ_1 and λ_2 when $\lambda_2 > 0.01$. Note that, when both λ_1 and λ_2 vary from 0.0001 to 0.005, the AUC performances significantly decrease especially on the Gulfport and San Diego datasets, which shows the importance of both spectral and spatial reconstruction terms. The optimal combination of λ_1 and λ_2 on the HYDICE Urban, Cat Island, Gulfport, San Diego, and EI Segundo datasets is (10,0.01), (0.05,5), (10,0.0001), (10,0.05), and (10,0.001), respectively.

4) Impact of p and q: The spectral feature dimension p and the spatial feature dimension q extracted by the two-stream autoencoders influence the feature learning capability of the network and thus influence the detection performance. Hence, we set $p \in \{4, 8, 16, 32, 64\}$ and $q \in \{7, 32, 128, 256, 512\}$ on each dataset to evaluate their impacts on the detection performance. As shown in Fig. 12, for the HYDICE Urban, San Diego, and EI Segundo datasets, when q increases, the AUC scores tend to decrease at different values of p. Hence, spectral information contributes more than spatial information to the optimal detection result on these datasets. On the contrary, when q increases, the AUC scores achieved on the Cat Island dataset tend to increase at different values of p. Moreover, if q is 512, the AUC scores are insensitive to p. Therefore, spatial information contributes more than spectral information to the optimal detection result on this dataset. Note that the AUC scores are insensitive to p and q on the Gulfport dataset and remain higher than 99.00%. The optimal combination of p and q on the HYDICE Urban, Cat Island, Gulfport, San Diego, and EI Segundo datasets is (32,7), (64,512), (32,7), (32,7), and (32,7), respectively.

5) Impact of w: Since the spatial window size w of the image patch centered at each pixel plays an important role in spatial feature extraction [66], we investigate its impact on the detection results by setting $w \in \{3, 5, 7, 9, 11, 13, 15\}$. The detection results on the five datasets are presented in Fig. 13. It can be observed that the AUC scores remain at a high level with respect to w on the HYDICE Urban, Gulfport, and EI Segundo datasets. The highest AUC scores are achieved when w is 9, 9, and 5 on these three datasets. In contrast, the detection performance achieved on the San Diego dataset is high when w is at a low range. On the contrary, the detection performance achieved on the Cat Island dataset is high when w is at a high range. The optimal values of w are 5 and 9 on these two datasets. Note that the AUC scores tend to decline when w is larger than its optimal value on all datasets. This is because more noise or anomaly targets will be introduced in the image patch as the window size increases.

G. Running Time

We investigate the running time of different detectors. All methods are implemented in Python 3.7 on a PC with a 3.0 GHz Intel i7 CPU and 64-GB memory. The network is trained on an Nvidia GeForce RTX 2080 Ti GPU. As reported in Table VI, it is obvious that the LRX [34] method takes the highest time among the conventional detection methods on all

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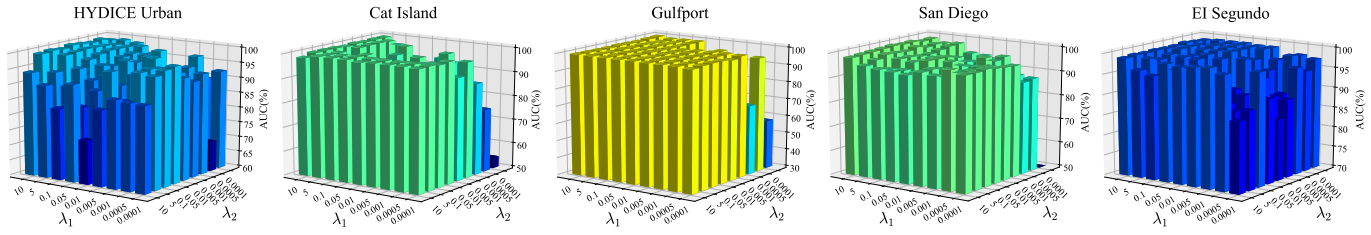


Fig. 11. AUC scores with respect to different settings of λ_1 and λ_2 on five datasets.

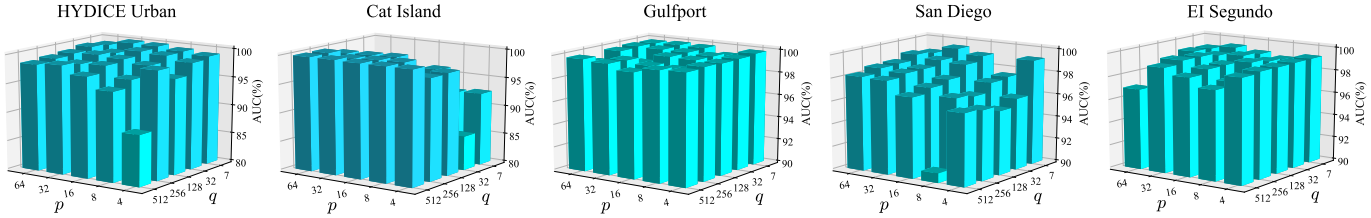


Fig. 12. AUC scores with respect to different settings of p and q on five datasets.

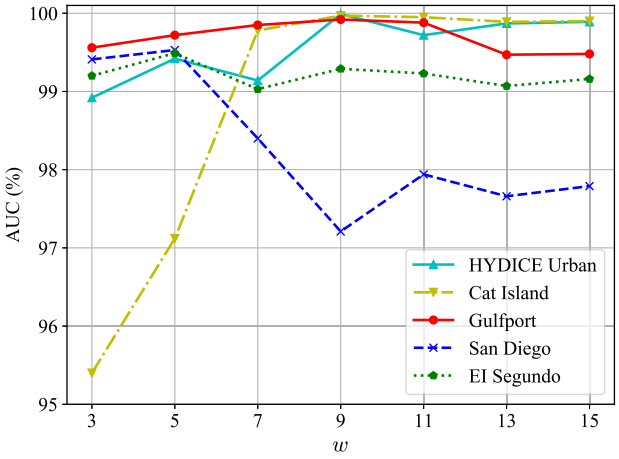


Fig. 13. AUC scores with respect to different settings of w on five datasets.

the datasets since it uses a pixelwise sliding dual window to estimate the background statistics, whereas the QLRX [64] method spends the least time with specialized optimization. Note that the running time of the deep learning-based DAEAD method [15] is comparable to that of the LRX method. In contrast, ss-DSVDD takes more running time than other detectors as it takes most computational time on the iterative feature learning process. Although QLRX [64], SVDD [7], CRD [43], SR-LMM [10], and LRASR [14] methods are efficient, they achieve lower accuracies than the proposed method.

V. Conclusion

In this article, we propose an end-to-end trainable deep one-class classification network named ss-DSVDD for HAD. By modeling the background with a compact hypersphere, the proposed method can detect anomalies without any statistic constraints on the latent representations. Moreover, the proposed method incorporates band selection, spectral–spatial feature learning, and anomaly detection into a united framework. In this way, the band selection reduces the negative effect of redundant bands on the detection performance. Meanwhile, the extracted spectral–spatial features are well-suited to anomaly detection. Experimental results validate the effectiveness of the proposed method and demonstrate its superiority compared to several state-of-the-art HAD methods.

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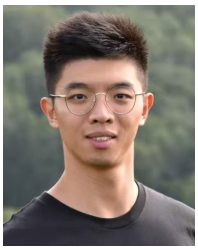
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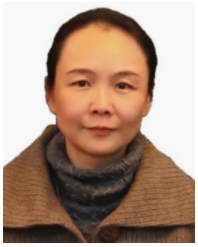
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